

Central Limit Theorem for k -tuples Frequencies in Mutation Systems

Liav Koram, Ohad Elishco¹

¹Ben-Gurion University of the Negev*

April 17, 2026

Abstract

Motivated by the high capacity and confidentiality of in-vivo DNA storage, we address the challenge of data degradation caused by biological mutations. We analyze the long-term behavior of such systems by investigating the evolution of k -tuple frequencies. Extending prior literature on mutation systems, we derive the asymptotic distribution of these frequencies. Our results offer a deeper understanding of the mutation process, facilitating the development of improved error-correction schemes for biological data storage.

Key Words : Coding theory, Capacity, Mutation systems, DNA storage

The exponential growth in global data generation has compelled researchers to look beyond traditional silicon and magnetic media, identifying DNA-based storage as a premier candidate for next-generation archival solutions. Leveraging the high information density, longevity, and sustainability of DNA molecules [Bal12, WkWF03, YGM17], this technology has sparked intensive interdisciplinary research spanning information theory, molecular biology, and coding theory. Generally, DNA storage systems are investigated along two primary trajectories. The first, in-vitro storage, operates in controlled laboratory environments where digital data is converted into nucleotide sequences, chemically synthesized, and stored for long-term preservation. Research in this domain focuses on optimizing encoding and synthesis processes [LLR⁺20, EH23, ASLY23, MRRY22, JFSB20], while simultaneously addressing error correction. These errors include insertions, deletions, and substitutions arising during synthesis or degradation [BGH⁺16, DM07, EZ17, LHT22, NCIK21, CCG⁺21, NCIK20, LSWZY19, YZWZ21], as well as read-errors introduced by sequencing technologies [BPRS20, MDSG16,

*elishco@gmail.com, liavk@post.bgu.ac.il

SSA⁺21, EGYM21, MY21]. While practically feasible, in-vitro methods face ongoing challenges regarding synthesis costs and throughput. The second trajectory, in-vivo storage, represents a biologically integrated paradigm where information is embedded within an organism’s genomic DNA. This allows data to persist and replicate via natural cellular processes. Although biological constraints and the necessity of preserving genetic function often result in lower information density compared to in-vitro methods, in-vivo storage offers distinct advantages. These include enhanced confidentiality through biological obscurity (steganography) [CRB99], genetic watermarking, and resilience against physical destruction [HB07, LDB⁺12, SNMC17, JFS⁺10]. Moreover, this paradigm enables multi-generational preservation and dynamic systems that leverage native cellular mechanisms. However, in-vivo storage is fundamentally challenged by the organism’s susceptibility to natural mutations. Unlike in-vitro systems, where errors occur at discrete stages (synthesis, storage, reading) and deletions typically reduce sequence length, in-vivo systems are subject to continuous evolutionary pressures. Biological processes introduce tandem duplications, varying-length deletions, and nucleotide substitutions. These mechanisms often increase sequence length and fundamentally alter composition over time. Consequently, maintaining data integrity across generations requires navigating the same selective pressures and random drift that shape natural genomes. Theoretical frameworks for mutation systems—encompassing duplication, substitution, and combined models—have been extensively studied. Existing literature has characterized their fundamental properties [JFB17, BTS22], analyzed capacity limits [LSBF19, EHSB19, FSB15], and developed error-correcting codes to mitigate mutational effects [KT18, Kov22, LJWZ18, LWZY17, TLF21, LWZY19, JFSB17, MV17, SGSD17, TLF21, TYSF20]. These advances establish the limits of reliable information storage under iterative mutation. This paper builds upon [Eli24], which established the almost-sure convergence of k -tuple frequencies to deterministic limits in mutation systems. Here, we extend that work by deriving the asymptotic probability distribution of these frequencies. By characterizing the precise limiting distributions and the statistical properties of fluctuations around the equilibrium, we provide a deeper mathematical understanding of long-term mutation behavior. These theoretical insights have significant practical implications for designing robust error-correction strategies and optimal reconstruction algorithms tailored specifically for the dynamic environment of in-vivo DNA storage.

1 preliminaries

Let $\mathcal{A} = \{a_0, \dots, a_{d-1}\}$ be a finite alphabet of size $|\mathcal{A}| = d$. A word (or a string) $w = w_0w_1 \dots w_{n-1}$ over $\mathcal{A}$ is a finite concatenation of symbols from $\mathcal{A}$. We use the notation $|\cdot|$ to indicate both set size and word length.

The set $\mathcal{A}^k$ denotes all k -length words (also called k -tuples) over $\mathcal{A}$, and $\mathcal{A}^* := \bigcup_{m=1}^{\infty} \mathcal{A}^m$ is the set of all finite words over $\mathcal{A}$.

For a set $A \subseteq \mathbb{Z}$ and $n \in \mathbb{Z}$ we write $n + A := \{n + j : j \in A\}$. We also use $[n]$ to denote the set $[n] := \{0, 1, \dots, n-1\}$. For a word $w \in \mathcal{A}^n$ and for $A \subseteq [n]$, we denote by $w_{[A]}$ the restriction of w to the positions in A . For $0 \leq i < j < n$, we also write w_i^j to denote $w_i^j = w_{i+[j-i+1]} = w_i w_{i+1} \dots w_j$.

For finite words $u, v \in \mathcal{A}^*$, we denote by uv the concatenation of u and v , and by u^n the concatenation of u with itself n times. We say that u is a **subword** of v , if $|u| \leq |v|$ and there exists $i \in [|v|]$ such that $v_{i+[|u|]} = u$, with coordinates taken modulo $|v|$.

Throughout, we use the indicator function, defined by $\mathbb{1}_{[a=b]} = \begin{cases} 1 & \text{if } a = b \\ 0 & \text{else} \end{cases}$.

We use $\text{ct}_v(u)$ to denote the number of times u appears in v as a subword, i.e., $\text{ct}_v(u) := \sum_{i=0}^{|v|} \mathbb{1}_{[u=v_{i+[|u|]}]}$ where coordinates are taken modulo $|v|$. We also define a notation for the empirical probability of u in v ,

$$\text{fr}_v(u) = \begin{cases} 0 & \text{if } |u| > |v| \\ \frac{1}{|v|} \text{ct}_v(u) & \text{if } |u| \leq |v| \end{cases}.$$

We often use vectors to denote the number of appearances and the empirical probability. For a given $k \in \mathbb{N}$, the count vector $\text{ct}_v^{(k)} \in \mathbb{R}^{d^k}$ has coordinates corresponding to k -tuples. For any $u \in \mathcal{A}^k$, the u coordinate is defined as $\text{ct}_v^{(k)}(u)$. When k is clear from the context, we omit it. Similarly, $\text{fr}_v^{(k)}$ denotes the vector of frequencies. We now define a mutation law.

Definition 1.1. Let $\mathcal{A} = \{a_0, \dots, a_{d-1}\}$ represent a finite alphabet with a size of $|\mathcal{A}| = d$. A **mutation law** $\vartheta : \mathcal{A} \rightarrow \mathcal{A}^*$ is a (random) function with domain $\mathcal{A}$. For every symbol $a \in \mathcal{A}$, $\vartheta(a)$ is a random word chosen according to a fixed (and known), finitely supported distribution p_a over $\mathcal{A}^*$. The distribution p_a could depend on the symbol a .

We say that ϑ is an **average τ -mutation law** if for every $t \in [d]$, $\mathbb{E}[|\vartheta(a_t)|] = \tau$.

Example 1.2. A mutation law is best explained by an example. Let $\mathcal{A} = \{0, 1\}$ be the binary alphabet, define a mutation law ϑ as follows

$$\vartheta(0) = \begin{cases} 1 & w.p. \frac{1}{2} \\ 00 & w.p. \frac{1}{2} \end{cases}, \quad \vartheta(1) = \begin{cases} 0 & w.p. \frac{1}{2} \\ 11 & w.p. \frac{1}{2} \end{cases}.$$

We stress that in our definition, a symbol cannot be deleted, but must be replaced with a word that contains at least one symbol. This implies that for every symbol $a \in \mathcal{A}$, $|\vartheta(a)| \geq 1$ with probability 1. We now define a mutation step (or mutation round).

Definition 1.3. Let $w = w_0 \dots w_{m-1} \in \mathcal{A}^*$ be a word over an alphabet $\mathcal{A}$ with $|w| = m$. Let ϑ be a mutation law over $\mathcal{A}$. A **mutation step** (or mutation round) is performed by first selecting a position $i \in [m]$ uniformly at random, and then applying ϑ on w_i to obtain the random word $w_0 \dots w_{i-1} \vartheta(w_i) w_{i+1} \dots w_{m-1}$. With a slight abuse of notation, we denote the obtained (random) word by $\vartheta(w)$.

Example 1.4. We continue with the mutation law given in Example 1.2. It is straightforward to check that the mutation law is an average $(3/2)$ -mutation law. If we start with $w = 01$, then we can calculate the probability that $\vartheta(w) = 11$. The only way we can obtain 11 after a single mutation step is if 0 is selected and then transformed into 1. Therefore, we choose the first symbol in w in the mutation step (probability $1/2$), and then the symbol is transformed into 1 (probability $1/2$). Overall, $\Pr(\vartheta(w) = 11) = 1/4$. Determining whether and how one word can mutate into another is generally straightforward, making the calculation of this probability equally tractable.

The next definition encapsulates the evolution of a word under repeated mutations.

Definition 1.5. A **mutation system** is defined by $S = (\mathcal{A}, w, \vartheta)$, where $\mathcal{A}$ is a finite alphabet, $S(0) = w \in \mathcal{A}^*$ is a non-empty starting word, and ϑ is a mutation law defined over $\mathcal{A}$. We call S an **average τ -mutation system** if ϑ is an average τ -mutation law. A mutation system evolves according to a discrete random model defined recursively. For $n \in \mathbb{N}$, set $S(n) = \vartheta(S(n-1)) = \vartheta^n(w)$, where $\vartheta^n(w)$ is the random word obtained by application of n mutation steps, starting with the word w .

A matrix $\mathbf{M} \in \mathcal{M}^{n \times m}$ is called **non-negative** if $\mathbf{M} \geq \mathbf{0}$ entry-wise. We use $\rho(\mathbf{M})$ to denote the spectral radius of $\mathbf{M}$, which is defined as the maximum absolute value (modulus) of its eigenvalues.

A non-negative matrix $\mathbf{M} \in \mathcal{M}^{n \times n}$ is called **irreducible** iff $(\mathbf{I} + \mathbf{M})^{n-1} > \mathbf{0}$ entry-wise. The following theorem states several known properties of irreducible non-negative matrices.

Definition 1.6. Let S be an average τ -mutation system over the alphabet $\mathcal{A} = \{a_0, \dots, a_{d-1}\}$ of size $|\mathcal{A}| = d$, and let $k \in \mathbb{N}$. The **k -substitution matrix** $\mathbf{M}^{(k)} := \mathbf{M}_{\vartheta}^{(k)}$ is a real $d^k \times d^k$ matrix, with non-negative entries, defined as follows. For $u = (u_0 u_1 \dots u_{k-1}), v = (v_0 v_1 \dots v_{k-1}) \in \mathcal{A}^k$, the (u, v) entry in $\mathbf{M}^{(k)}$ is

$$\begin{aligned} \mathbf{M}_{u,v}^{(k)} := & \sum_{l \in \mathbb{N}} \sum_{\eta \in \mathcal{A}^l} \sum_{t \in [d]} \Pr(\vartheta(a_t) = \eta) \left(\sum_{j=1}^{k-1} \mathbb{1}_{[v_j=a_t]} \mathbb{1}_{[(v_0^{j-1} \eta v_{j+1}^{k-1})]_{[k]}=u} \right. \\ & \left. + \sum_{j=0}^{l-1} \mathbb{1}_{[v_0=a_t]} \mathbb{1}_{[\eta_j^{l-1} v_1^{k-l+j}=u]} \right) \end{aligned}$$

where $(v_0^{j-1}\eta v_{j+1}^{k-1})_{[k]}$ is the sequence obtained by taking the first k symbols from $v_0^{j-1}\eta v_{j+1}^{k-1}$. When k is clear from the context, we write $\mathbf{M}$ instead of $\mathbf{M}^{(k)}$.

In [Eli24], it is shown that if ϑ is an average τ -mutation law, then $\rho(\mathbf{M}^{(k)}) = \tau + k - 1$. The main result of [Eli24] follows.

Theorem 1.7. [Eli24, Theorem IV.11] *Let S be an average τ -mutation system over the alphabet $\mathcal{A} = \{a_0, \dots, a_{d-1}\}$. Fix k and let w be a starting word of length $m \geq k$. Assume that the k -substitution matrix $\mathbf{M}^{(k)}$ has a unique real eigenvalue $\lambda_0 = \rho(\mathbf{M}^{(k)}) = \tau + k - 1$. Let $\mathbf{r}$ represent the right eigenvector associated with λ_0 , normalized to a probability vector. If the real part of every non-regular eigenvalue λ (geometric multiplicity is strictly smaller than the algebraic multiplicity) is strictly smaller than k , then $\mathbf{fr}_{S(n)} \xrightarrow{\text{Pr}} \mathbf{r}$.*

2 Main Results

In this section we present our main results. First, we may strengthen Theorem 1.7 by directly applying Lebesgue-Vitali's Theorem, obtaining the following corollary.

Corollary 2.1. *With the same notation and assumptions as in Theorem 1.7, we have $\mathbf{fr}_n^{(k)} \xrightarrow{L_1} \mathbf{r}$.*

Example 2.2. *Continuing with Example 1.2, we can calculate the substitution matrix for the system $S = (\mathcal{A}, w, \vartheta)$ for $k = 2, 3$, and obtain*

$$\mathbf{M}^{(2)} = \begin{bmatrix} \frac{3}{2} & 1 & \frac{1}{2} & 0 \\ \frac{1}{2} & 1 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 1 & \frac{1}{2} \\ 0 & \frac{1}{2} & 1 & \frac{3}{2} \end{bmatrix}, \quad \mathbf{M}^{(3)} = \begin{bmatrix} 2 & \frac{3}{2} & \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & 1 & \frac{1}{2} & 1 & 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{2} & 0 & 1 & \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 1 & \frac{3}{2} & 0 & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & 0 & \frac{3}{2} & 1 & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} & 1 & 0 & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 0 & 1 & \frac{1}{2} & 1 & \frac{1}{2} \\ 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & \frac{3}{2} & 2 \end{bmatrix}.$$

Notice that both $\mathbf{M}^{(2)}$ and $\mathbf{M}^{(3)}$ are irreducible. The right leading eigenvectors (normalized to be probability vectors) are $\mathbf{r}^{(2)} = \frac{1}{10}[3 \ 2 \ 2 \ 3]^T$, and $\mathbf{r}^{(3)} = \frac{1}{60}[11 \ 7 \ 5 \ 7 \ 7 \ 5 \ 7 \ 11]^T$. From 2.1, the asymptotic frequency of pairs is given by $\mathbf{fr}_{S(n)}^{(2)} \rightarrow \mathbf{r}^{(2)}$ and of triples is given by $\mathbf{fr}_{S(n)}^{(3)} \rightarrow \mathbf{r}^{(3)}$ in L_1 .

Let $\mathbf{M} = \mathbf{M}^{(k)}$ be a substitution matrix for some fixed k . For an eigenvalue λ_i of $\mathbf{M}$, we denote by $\mathbf{u}^{(i)}$ the left eigenvector that corresponds to λ_i .

We express $\mathbf{u}^{(i)}$ and λ_i by their real and imaginary parts $\mathbf{u}^{(i)} = \mathbf{u}_r^{(i)} + \mathbf{i}\mathbf{u}_c^{(i)}$ and $\lambda_i = \lambda_{i,r} + \mathbf{i}\lambda_{i,c}$, respectively.

For $n \in \mathbb{N}$, let $\mathbf{ct}_n = \mathbf{ct}_{S(n)}^{(k)}$ be the count vector of k -tuples after the n th step. We define $X_n^{(i)}$ as the projection $X_n^{(i)} = \mathbf{u}^{(i)} \cdot \mathbf{ct}_n$, and we express $X_n^{(i)}$ also by its real and imaginary parts

$$X_n^{(i)} = X_n^{r(i)} + \mathbf{i}X_n^{c(i)} = \mathbf{u}_r^{(i)} \cdot \mathbf{ct}_n + \mathbf{i}\mathbf{u}_c^{(i)} \cdot \mathbf{ct}_n.$$

Throughout the paper, we assume that $\mathbf{M}$ is diagonalizable (the non-diagonalizable case will be discussed at the end).

Let $\lambda_0 = \rho(\mathbf{M})$ be the (unique) principal eigenvalue, i.e., the Perron eigenvalue. Since $\mathbf{M}$ has dimension $d^k \times d^k$, there are d^k eigenvalues. Since the matrix is real and since by assumptions the matrix is diagonalizable, complex eigenvalues come in conjugate pairs. Out of the $d^k - 1$ non-principal eigenvalues, suppose $2m$ are complex pairs $\lambda_{i+m} = \bar{\lambda}_i$, and the rest $d^k - 1 - 2m$ eigenvalues are purely real.

Definition 2.3. We define the vector for all $X_n^{r(i)}$ and $X_n^{c(i)}$ of all non-principal eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_{d^k-1}$ as

$$\mathbf{X}_n = \left[X_n^{r(1)}, X_n^{c(1)}, \dots, X_n^{r(m)}, X_n^{c(m)}, X_n^{r(2m+1)}, X_n^{r(2m+2)}, \dots, X_n^{r(d^k-1)} \right].$$

Notice the first $2m$ values are for the complex eigenvalue pairs, and the rest are for the purely real eigenvalues, that $\mathbf{X}_n$ contains only real values, and that in $\mathbf{X}_n$ we consider only one eigenvalue from each complex pair. We are now ready to state our main result.

Theorem 2.4. Let S be an average τ -mutation system with irreducible substitution matrix for k -tuples $\mathbf{M}^{(k)}$, and assume that the frequency of $(2k-1)$ -tuples converges in probability to some positive vector $\mathbf{fr}^{(2k-1)} \xrightarrow{P} \mathbf{r}^{(2k-1)} > 0$. If for every non-leading eigenvalue $\lambda = \lambda_r + \mathbf{i}\lambda_c$, we have $\frac{\lambda_r - k}{\tau - 1} < \frac{1}{2}$, then $n^{-\frac{1}{2}}\mathbf{X}_n$ has an asymptotic joint normal distribution.

Recall that a vector has a joint normal distribution iff every linear sum of its entries has a normal distribution. Thus, it is equivalent to say that for every set of real coefficients $\alpha_r^{(i)}, \alpha_c^{(i)} \in \mathbb{R}$, assuming at least one coefficient is non-zero, the sum

$$\frac{1}{\sqrt{n}} \sum_{i=1}^m \left(\alpha_r^{(i)} X_n^{r(i)} + \alpha_c^{(i)} X_n^{c(i)} \right) + \frac{1}{\sqrt{n}} \sum_{i=2m+1}^{p-1} \alpha_r^{(i)} X_n^{r(i)}$$

converges in distribution as $n \rightarrow \infty$ to a random variable Z , with a characteristic function $\mathbb{E}[e^{(-\frac{1}{2}\sigma^2 t^2)}]$, where σ^2 is a non-negative constant. In case $\sigma^2 = 0$, the sum converges to 0 in probability. Later on, we show how to find σ^2 and find a condition to ensure $\sigma^2 > 0$.

Example 2.5. Consider our running example as in Example 1.2 and fix $k = 2$. Recall that the mutation law is a $3/2$ -average mutation law. Since $\mathbf{M}^{(3)}$ is irreducible, Theorem 1.7 implies that $\mathbf{fr}^{(2k-1)}$ converges in probability to a positive vector $\mathbf{r}^{(2k-1)} = \mathbf{r}^{(3)} > 0$. Moreover, $\mathbf{M}^{(2)}$ is also irreducible, and the eigenvalues of $\mathbf{M}^{(2)}$ are

$$\lambda_0 = \frac{5}{2}, \lambda_1 = \frac{3}{2}, \lambda_2 = 1, \lambda_3 = 0.$$

Notice that $\lambda_0 = \tau + k - 1 = 3/2 + 2 - 1 = 5/2$ and that every non-leading eigenvalue satisfies the inequality $\frac{\lambda_r - k}{\tau - 1} < \frac{1}{2}$. The left eigenvectors of $\mathbf{M}^{(2)}$ are

$$\begin{aligned} \mathbf{u}_0 &= [1 \quad 1 \quad 1 \quad 1], \quad \mathbf{u}_1 = [-1 \quad -1 \quad 1 \quad 1], \\ \mathbf{u}_2 &= [0 \quad -1 \quad 1 \quad 0], \quad \mathbf{u}_3 = [1 \quad -\frac{3}{2} \quad -\frac{3}{2} \quad 1]. \end{aligned}$$

From Theorem 2.4, the projection vector $\mathbf{X}_n$ of $\mathbf{ct}_{S(n)}^{(2)}$ on $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ satisfies

$$\lim_{n \rightarrow \infty} n^{-\frac{1}{2}} \mathbf{X}_n \sim \mathcal{N}(0, \Sigma).$$

We later show how to calculate σ^2 in $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \sum_{i=1}^3 \alpha_i \mathbf{u}_i \cdot \mathbf{ct}_{S(n)}^{(2)} \sim \mathcal{N}(0, \sigma^2)$, for every set of α_i .

Overall, throughout the paper, we assume that S is an average τ -mutation, $\tau > 1$, the matrix $\mathbf{M}^{(k)}$ is irreducible and has a regular eigendecomposition ($\mu_a(\lambda) = \mu_g(\lambda)$ for every eigenvalue λ) unless stated otherwise, we assume that $\mathbf{fr}^{(2k-1)}$ converges in probability to some positive vector $\mathbf{r}^{(2k-1)}$, and that for every non-leading eigenvalue $\lambda = \lambda_r + \mathbf{i}\lambda_c$, we have $\frac{\lambda_r - k}{\tau - 1} < \frac{1}{2}$.

3 Walkthrough of the Proof

In this section we will showcase the main steps of the proof of 2.4 giving an overview of the proof and showing how to calculate the variance σ^2 .

The general idea of the proof is to construct a sequence that is asymptotically equivalent to a **martingale difference** sequence and captures the behavior of the system, and those sequences will be easier to analyze. We follow a similar approach to that of [Eli24], using analogous notation and relying on several results established therein. We restate the relevant definitions and results from [Eli24].

Throughout this section, we will drop the i index from expressions like $\mathbf{u}_r^{(i)}$ and $\mathbf{X}_n^{(i)}$, as the rest true for all i . To capture the change in the k -tuples composition in step n , we define $\nabla \mathbf{ct}_n^{(k)} := \mathbf{ct}_n^{(k)} - \mathbf{ct}_{n-1}^{(k)}$. Similarly, we define

the following

$$\begin{aligned}\nabla X_n^r &:= X_n^r - X_{n-1}^r = \mathbf{u}_r \cdot \nabla \mathbf{c} \mathbf{t}_n, & \nabla X_n^c &:= X_n^c - X_{n-1}^c = \mathbf{u}_c \cdot \nabla \mathbf{c} \mathbf{t}_n \\ M_n^r &:= \nabla X_n^r - \frac{1}{Y_{n-1}} ((\lambda_r - k)X_{n-1}^r - \lambda_c X_{n-1}^c), \\ M_n^c &:= \nabla X_n^c - \frac{1}{Y_{n-1}} ((\lambda_r - k)X_{n-1}^c + \lambda_c X_{n-1}^r),\end{aligned}$$

where Y_n is the length of the mutation system in step n , $Y_n = |S(n)|$. We denote by $\mathcal{F}_n$ the σ -algebra generated by $\mathbf{c} \mathbf{t}_j$, $j \leq n$. In [Eli24, Claim IV.13] the following claim is proved.

Claim 3.1 (proved in [Eli24]). M_n^r and M_n^c are martingale differences, with respect to $\{\mathcal{F}_j\}_j$, with zero mean $E[M_n^r] = E[M_n^c] = 0$.

While the processes M_n^r, M_n^c are martingale, the random length Y_n makes them difficult to study. Similarly to [Eli24], we approximate the processes M_n by replacing Y_n with its mean $E[Y_n] = n\tau + |w|$ with $|w|$ being the length of the initial word w . Clearly, the obtained processes are no longer martingale. Define

$$\overline{M}_n^r := \nabla X_n^r - \frac{(\lambda_r - k)X_{n-1}^r - \lambda_c X_{n-1}^c}{(n-1)(\tau-1) + |w|}, \quad \overline{M}_n^c := \nabla X_n^c - \frac{(\lambda_r - k)X_{n-1}^c + \lambda_c X_{n-1}^r}{(n-1)(\tau-1) + |w|}.$$

To claim that we may analyze $\overline{M}_n^r, \overline{M}_n^c$ instead of M_n^r, M_n^c , we show asymptotic equivalence (the proof is omitted due to page restrictions).

Claim 3.2. $\overline{M}_n^r, \overline{M}_n^c$ are asymptotically equivalent to M_n^r, M_n^c under L_1 , respectively, i.e. $\lim_{n \rightarrow \infty} \mathbb{E} [|\overline{M}_n^r - M_n^r|] = 0$, $\lim_{n \rightarrow \infty} \mathbb{E} [|\overline{M}_n^c - M_n^c|] = 0$. We denote this asymptotic equivalence by $\overline{M}_n^r \stackrel{L_1}{\sim} M_n^r$ and $\overline{M}_n^c \stackrel{L_1}{\sim} M_n^c$.

Moving along similar lines as [Eli24], for fixed $\alpha_r, \alpha_c \in \mathbb{R}$, we seek constants $\beta_{j,n}^r, \beta_{j,n}^c$ for $j = 1, \dots, n$ such that

$$\sum_{j=1}^n (\beta_{j,n}^r \overline{M}_j^r + \beta_{j,n}^c \overline{M}_j^c) = \alpha_r X_n^r + \alpha_c X_n^c + \varepsilon_n,$$

where $\varepsilon_n = O\left(n^{\frac{\lambda_r - k}{\tau - 1}}\right)$ is a small error term. In the case that the corresponding left eigenvector $\mathbf{u}_i$ is purely real, $X_n^c = 0$, so we will set α_c to be 0 as well.

Finding $\beta_{j,n}^r, \beta_{j,n}^c$ is a lengthy and straightforward process, and therefore will be omitted. We will mention that their values can be bounded above by

$$|\beta_{j,n}^r|, |\beta_{j,n}^c| \leq (|\alpha_r| + |\alpha_c|) \left(\frac{n}{j}\right)^{\frac{\lambda_r - k}{\tau - 1}} + O\left(n^{\frac{\lambda_r - k}{\tau - 1} - 1}\right).$$

Next, we define the following sums,

$$v_n := \sum_{j=1}^n (\beta_{j,n}^r M_j^r + \beta_{j,n}^c M_j^c)$$

$$\bar{v}_n := \sum_{j=1}^n (\beta_{j,n}^r \bar{M}_j^r + \beta_{j,n}^c \bar{M}_j^c) = \alpha_r X_n^r + \alpha_c X_n^c + \varepsilon_n,$$

which allow us to approximate the linear sums of X_n^r and X_n^c by sums of martingale differences through the following claim (the proof is omitted due to page restriction).

Claim 3.3. *Let $\lambda = \lambda_r + \mathbf{i}\lambda_c$ be a non-leading eigenvalue with $\frac{\lambda_r - k}{\tau - 1} < \frac{1}{2}$. Let v_n and $\bar{v}_n$ be the sums corresponding to λ . Then $\frac{\bar{v}_n}{\sqrt{n}} \stackrel{L_1}{\approx} \frac{v_n}{\sqrt{n}}$.*

The proof of Claim 3.3 is removed due to page restriction.

From the previous claims, by summing over all $\bar{v}_n$ of all non-leading eigenvectors λ_i , $i = 1, \dots, d^k - 1$ of $\mathbf{M}^{(k)}$, we obtain that

$$\frac{1}{\sqrt{n}} \left(\sum_{i=1}^m (\alpha_r^{(i)} X_n^{r(i)} + \alpha_c^{(i)} X_n^{c(i)}) + \sum_{i=2m+1}^{d^k-1} \alpha_r^{(i)} X_n^{r(i)} \right)$$

is asymptotically equivalent to $\frac{1}{\sqrt{n}} \left(\sum_{i=1}^m v_n^{(i)} + \sum_{i=2m+1}^{d^k-1} v_n^{(i)} \right)$.

We notice the small error term ε_n was removed because by dividing in $\sqrt{n}$ the error converges to 0. $\varepsilon_n/\sqrt{n} = O\left(n^{\frac{\lambda_r - k}{\tau - 1}}\right)/\sqrt{n} \rightarrow 0$.

Our goal now is use martingale CLT for v_n and use the CLT for martingales to prove the main theorem.

We define $\mathcal{F}_{n,j}$ as the σ -algebra generated by $\mathbf{ct}_l$ for $l \leq j$, and

$$\mathcal{X}_{n,j} = \frac{1}{\sqrt{n}} \left(\sum_{i=1}^m (\beta_{j,n}^{r(i)} M_j^{r(i)} + \beta_{j,n}^{c(i)} M_j^{c(i)}) + \sum_{i=2m+1}^{d^k-1} \beta_{j,n}^{r(i)} M_j^{r(i)} \right).$$

Clearly, since $M_j^{r(i)}$ and $M_j^{c(i)}$ are martingale differences for every i , their sum over i is also a martingale difference.

In order to use the martingale CLT (see, for example [HH14, Th. 3.2 and Cor. 3.1]), we show that $\mathcal{X}_{n,j}$ satisfy the conditional Lindeberg condition and the conditional variance condition. Again, due to page limitations, we omit the proofs.

From the martingale CLT, we obtain that

$$\frac{1}{\sqrt{n}} \sum_{i=1}^m (\alpha_r^{(i)} X_n^{r(i)} + \alpha_c^{(i)} X_n^{c(i)}) + \frac{1}{\sqrt{n}} \sum_{i=2m+1}^{d^k-1} \alpha_r^{(i)} X_n^{r(i)}$$

converges in distribution to a normally distributed random variable Z with 0 mean and a variance of σ^2 . This follows since

$$\begin{aligned} & \frac{1}{\sqrt{n}} \sum_{i=1}^m \left(\alpha_r^{(i)} X_n^{r(i)} + \alpha_c^{(i)} X_n^{c(i)} \right) + \frac{1}{\sqrt{n}} \sum_{i=2m+1}^{d^k-1} \alpha_r^{(i)} X_n^{r(i)} \\ & \stackrel{L^1}{\sim} \frac{1}{\sqrt{n}} \sum_{j=1}^n \left(\sum_{i=1}^m \left(\beta_{j,n}^{r(i)} M_j^{r(i)} + \beta_{j,n}^{c(i)} M_j^{c(i)} \right) + \sum_{i=2m+1}^{d^k-1} \beta_{j,n}^{r(i)} M_j^{r(i)} \right) \\ & \xrightarrow{d} Z. \end{aligned}$$

We now show how to find σ^2 . Let $m_0, m_1, \dots, m_{d^k-1}$ be the lexicographic enumeration of all the k -tuples $\mathcal{A}^k$. For $i, j \in [d^k]$, define

$$\begin{aligned} \theta_{m_i, m_j} &:= \sum_{v \in \mathcal{A}^{2k-1}} \sum_{\eta \in \text{Supp}(\vartheta(v_{k-1}))} \mathbf{r}^{(2k-1)}(v) \Pr(\vartheta(v_{k-1}) = \eta) \\ & \left(\mathbf{ct}_{v_0^{k-2} \eta v_k^{2k-2}}^{(k)}(m_i) - \mathbf{ct}_v^{(k)}(m_i) \right) \left(\mathbf{ct}_{v_0^{k-2} \eta v_k^{2k-2}}^{(k)}(m_j) - \mathbf{ct}_v^{(k)}(m_j) \right). \end{aligned}$$

Notice that v is going over all the $(2k-1)$ -tuples and that η is going over the image of the mutation law ϑ applied on the central symbol in v , and $\mathbf{r}^{(2k-1)}$ is the leading right eigenvector of $\mathbf{M}^{(2k-1)}$.

We define the **Theta matrix** using θ_{m_i, m_j} as $\Theta := [\theta_{m_i, m_j}]_{i, j \in [d^k]}$.

Example 3.4. *Continuing with our running example, for $k = 2$, the Θ matrix for our mutation law is*

$$\Theta = \frac{1}{60} \begin{bmatrix} 61 & -16 & -16 & -14 \\ -16 & 16 & 16 & -16 \\ -16 & 16 & 16 & -16 \\ -14 & -16 & -16 & 61 \end{bmatrix}.$$

The matrix will be used to calculate σ^2 for any values of α_i s as we would see below.

Given Θ we can calculate σ^2 . Given the space constraints we will show the equation for the case where all eigenvalues of $\mathbf{M}^{(k)}$ are real. For $i \in [d^k]$, let $\mathbf{u}_i$ be the left eigenvector of $\mathbf{M}^{(k)}$ that corresponds to λ_i . We have that $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \sum_{i=1}^{d^k-1} \alpha_i X_n^{(i)} \sim N(0, \sigma^2)$, where $\sigma^2 = \sum_{i=1}^{d^k-1} \sum_{l=1}^{d^k-1} \frac{\alpha_i \alpha_l}{1 - \frac{\lambda_i + \lambda_l - 2k}{\tau - 1}} \mathbf{u}_i \Theta (\mathbf{u}_l)^T$.

We conclude by mentioning that although the derivation presented assumes the matrix $\mathbf{M}$ is diagonalizable, a similar result holds for the non-diagonalizable case. In addition, the resulting limiting frequencies are instrumental in bounding the capacity of mutation systems, serving as an increasingly tight approximation as k grows. This characterization provides a crucial first step toward the development of efficient error-correcting codes for mutation-based storage.

References

- [ASLY23] M. Abu-Sini, A. Lenz, and E. Yaakobi. DNA synthesis using shortmers. In *2023 IEEE International Symposium on Information Theory (ISIT)*, pages 585–590, 2023.
- [Bal12] Félix Balado. Capacity of DNA data embedding under substitution mutations. *IEEE Transactions on Information Theory*, 59(2):928–941, 2012.
- [BGH⁺16] M. Blawat, K. Gaedke, I. Huetter, X. M. Chen, B. Turczyk, S. Inverso, B. W. Pruitt, and G. M. Church. Forward error correction for DNA data storage. *Procedia Computer Science*, 80:1011–1022, 2016.
- [BPRS20] Vinnu Bhardwaj, Pavel A Pevzner, Cyrus Rashtchian, and Yana Safonova. Trace reconstruction problems in computational biology. *IEEE Transactions on Information Theory*, 67(6):3295–3314, 2020.
- [BTS22] E. Ben-Tolila and M. Schwartz. On the reverse-complement string-duplication system. *IEEE Transactions on Information Theory*, 68(11):7184–7197, 2022.
- [CCG⁺21] K. Cai, Y. M. Chee, R. Gabrys, H. M. Kiah, and T. T. Nguyen. Correcting a single indel/edit for DNA-based data storage: Linear-time encoders and order-optimality. *IEEE Transactions on Information Theory*, 67(6):3438–3451, 2021.
- [CRB99] C. T. Clelland, V. Risca, and C. Bancroft. Hiding messages in DNA microdots. *Nature*, 399(6736):533–534, 1999.
- [DM07] E. Drinea and M. Mitzenmacher. Improved lower bounds for the capacity of i.i.d. deletion and duplication channels. *IEEE Trans. on Inf. Theory*, 53(8):2693–2714, 2007.
- [EGYM21] O. Elishco, R. Gabrys, E. Yaakobi, and M. Médard. Repeat-free codes. *IEEE Transactions on Information Theory*, 67(9):5749–5764, 2021.
- [EH23] O. Elishco and W. Huleihel. Optimal reference for DNA synthesis. *IEEE Transactions on Information Theory*, 69(11):6941–6955, 2023.
- [EHSB19] O. Elishco, F. F. Hassanzadeh, M. Schwartz, and J. Bruck. The entropy rate of some Pólya string models. *IEEE Transactions on Information Theory*, 65(12):8180–8193, 2019.

- [Eli24] O. Elishco. On the long-term behavior of k -tuples frequencies in mutation systems. *IEEE Transactions on Information Theory*, 70(12):8524–8545, 2024.
- [EZ17] Y. Erlich and D. Zielinski. DNA fountain enables a robust and efficient storage architecture. *science*, 355(6328):950–954, 2017.
- [FSB15] F. Farnoud, M. Schwartz, and J. Bruck. The capacity of string-duplication systems. *IEEE Transactions on Information Theory*, 62(2):811–824, 2015.
- [HB07] D. Heider and A. Barnekow. DNA-based watermarks using the DNA-crypt algorithm. *BMC bioinformatics*, 8(1):1–10, 2007.
- [HH14] P. Hall and C. C. Heyde. *Martingale limit theory and its application*. Academic press, 2014.
- [JFB17] S. Jain, F. Farnoud, and J. Bruck. Capacity and expressiveness of genomic tandem duplication. *IEEE Transactions on Information Theory*, 63(10):6129–6138, 2017.
- [JFS⁺10] D. C. Jupiter, T. A. Ficht, J. Samuel, Q. M. Qin, and P. De Figueiredo. DNA watermarking of infectious agents: Progress and prospects. *PLoS pathogens*, 6(6):e1000950, 2010.
- [JFSB17] S. Jain, F. Farnoud, M. Schwartz, and J. Bruck. Duplication-correcting codes for data storage in the DNA of living organisms. *IEEE Transactions on Information Theory*, 63(8):4996–5010, 2017.
- [JFSB20] S. Jain, F. Farnoud, M. Schwartz, and J. Bruck. Coding for optimized writing rate in DNA storage. In *2020 IEEE International Symposium on Information Theory (ISIT)*, pages 711–716, 2020.
- [Kov22] M. Kovacevic. On the maximum number of non-confusable strings evolving under short tandem duplications. *Problems of Information Transmission*, 58:111–121, 2022.
- [KT18] M. Kovačević and V. Y. F. Tan. Asymptotically optimal codes correcting fixed-length duplication errors in DNA storage systems. *IEEE Communications Letters*, 22(11):2194–2197, 2018.
- [LDB⁺12] M. Liss, D. Daubert, K. Brunner, K. Kliche, U. Hammes, A. Leiherer, and R. Wagner. Embedding permanent watermarks in synthetic genes. 2012.

- [LHT22] Y. Liu, X. He, and X. Tang. Capacity-achieving constrained codes with GC-content and runlength limits for DNA storage. In *IEEE International Symposium on Information Theory (ISIT)*, pages 198–203, 2022.
- [LJWZ18] A. Lenz, N. Jünger, and A. Wachter-Zeh. Bounds and constructions for multi-symbol duplication error correcting codes. *arXiv preprint arXiv:1807.02874*, 2018.
- [LLR⁺20] Andreas Lenz, Yi Liu, Cyrus Rashtchian, Paul H. Siegel, Antonia Wachter-Zeh, and Eitan Yaakobi. Coding for efficient DNA synthesis. In *2020 IEEE International Symposium on Information Theory (ISIT)*, pages 2885–2890, 2020.
- [LSBF19] H. Lou, M. Schwartz, J. Bruck, and F. Farnoud. Evolution of k -mer frequencies and entropy in duplication and substitution mutation systems. *IEEE Transactions on Information Theory*, 66(5):3171–3186, 2019.
- [LSWZY19] A. Lenz, P. H. Siegel, A. Wachter-Zeh, and E. Yaakobi. Coding over sets for DNA storage. *IEEE Transactions on Information Theory*, 66(4):2331–2351, 2019.
- [LWZY17] A. Lenz, A. Wachter-Zeh, and E. Yaakobi. Bounds on codes correcting tandem and palindromic duplications. *arXiv preprint arXiv:1707.00052*, 2017.
- [LWZY19] A. Lenz, A. Wachter-Zeh, and E. Yaakobi. Duplication-correcting codes. *Designs, Codes and Cryptography*, 87:277–298, 2019.
- [MDSG16] A. Magner, J. Duda, W. Szpankowski, and A. Grama. Fundamental bounds for sequence reconstruction from nanopore sequences. *IEEE Trans. on Molecular, Biological and Multi-Scale Comm.*, 2(1):92–106, 2016.
- [MRRY22] K. Makarychev, M. Z. Rácz, C. Rashtchian, and S. Yekhanin. Batch optimization for DNA synthesis. *IEEE Transactions on Information Theory*, 68(11):7454–7470, 2022.
- [MV17] H. MahdaviFar and A. Vardy. Asymptotically optimal sticky-insertion-correcting codes with efficient encoding and decoding. In *2017 IEEE International Symposium on Information Theory (ISIT)*, pages 2683–2687. IEEE, 2017.
- [MY21] S. Marcovich and E. Yaakobi. Reconstruction of strings from their substrings spectrum. *IEEE Transactions on Information Theory*, 67(7):4369–4384, 2021.

- [NCIK20] T. T. Nguyen, K. Cai, K. A. S. Immink, and H. M. Kiah. Constrained coding with error control for DNA-based data storage. In *2020 IEEE International Symposium on Information Theory (ISIT)*, pages 694–699, 2020.
- [NCIK21] T. T. Nguyen, K. Cai, K. A. S. Immink, and H. M. Kiah. Capacity-approaching constrained codes with error correction for DNA-based data storage. *IEEE Transactions on Information Theory*, 67(8):5602–5613, 2021.
- [SGSD17] F. Sala, R. Gabrys, C. Schoeny, and L. Dolecek. Exact reconstruction from insertions in synchronization codes. *IEEE Transactions on Information Theory*, 63(4):2428–2445, 2017.
- [SNMC17] S. L. Shipman, J. Nivala, J. D. Macklis, and G. M. Church. CRISPR–Cas encoding of a digital movie into the genomes of a population of living bacteria. *Nature*, 547(7663):345–349, 2017.
- [SSA⁺21] R. Shafir, O. Sabary, L. Anavy, E. Yaakobi, and Z. Yakhini. Sequence reconstruction under stutter noise in enzymatic DNA synthesis. In *IEEE Inf. Theory Workshop (ITW)*, pages 1–6, 2021.
- [TLF21] Y. Tang, H. Lou, and F. Farnoud. Error-correcting codes for short tandem duplications and at most p substitutions. In *2021 IEEE International Symposium on Information Theory (ISIT)*, pages 1835–1840. IEEE, 2021.
- [TYSF20] Y. Tang, Y. Yehezkeally, M. Schwartz, and F. Farnoud. Single-error detection and correction for duplication and substitution channels. *IEEE Transactions on Information Theory*, 66(11):6908–6919, 2020.
- [WkWF03] P. C. Wong, K. k. Wong, and H. Foote. Organic data memory using the DNA approach. *Communications of the ACM*, 46(1):95–98, 2003.
- [YGM17] S. M. H. T. Yazdi, R. Gabrys, and O. Milenkovic. Portable and error-free DNA-based data storage. *Scientific reports*, 7(1):5011, 2017.
- [YZWZ21] Q. Yin, Y. Zheng, B. Wang, and Q. Zhang. Design of constraint coding sets for archive DNA storage. *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 19(6):3384–3394, 2021.